

Quartic Zero-Repair: Positive Packet Transport and Waring Phase Carriers

An incomplete preprint toward the logarithmic reserve bound

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Abstract

We study the exact history-indexed zero-repair problem for the explicit quartic pair

$$P = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = \frac{1}{64}(7, 20, 10, 20, 7),$$

whose probability generating functions satisfy

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

The cost $\mathcal{B}_n$ is the minimum expected initial nonnegative reserve required to transform fresh iid Q -inputs into iid P -outputs for n causal steps. We record a self-contained logarithmic lower bound, a weak- L^1 reduction of the matching upper bound, the exact one-fan stochastic-order characterization, and an exact tail-interlacing theorem that preserves the leading $1/j$ tail in all five children.

The main constructive contribution of this preprint is a positive three-regime Q -fresh packet fan. For every parent law with boundary mass $p = B(0) \leq 1/4$, a packet of mass $3p$ can compensate the boundary output defect while leaving public branches 1 and 3 exactly unchanged. A top-tail choice gives explicit low-prefix recursions and the sharp one-step boundary criterion

$$14B(0) + 4B(1) \leq 5.$$

We then pass to a Waring/Beta phase basis in which downward motion at the hard boundary has the exact aging law

$$DW_{0,k} = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}W_{0,k+1}.$$

This produces a positive phase carrier closed under the shift operators appearing in the packet fan. The remaining load-bearing statement is formulated as a Waring packet invariance conjecture. If it holds from the critical root $H_2 = W_{0,1}$, then

$$\mathcal{B}_n \leq 2H_{4n} = O(\log n),$$

and hence the logarithmic order is sharp.

This is intentionally an incomplete preprint: the all-history Waring packet invariance statement is not claimed as proved.

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1 The quartic zero-repair problem

Let

$$P = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = \frac{1}{64}(7, 20, 10, 20, 7). \quad (1)$$

Their probability generating functions obey

$$\boxed{G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}}, \quad (2)$$

and both laws have mean 2.

Definition 1.1 (Exact history-indexed tree). For every output history $h \in \{0, 1, 2, 3, 4\}^{\leq n}$, let B_h be a probability law on $\mathbb{N}_0$. The family is an exact depth- n zero-repair tree if at every internal node

$$\boxed{Q * B_h = \sum_{x=0}^4 P_x \delta_x * B_{hx}}. \quad (3)$$

For finite horizon we require the root to have finite first moment.

Definition 1.2 (Reserve bill). Let $\mathcal{K}_n$ denote the set of root laws of exact depth- n trees. Define

$$\boxed{\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}_B J, \quad \mathcal{B}_0 := 0.} \quad (4)$$

The convolution identity is not merely a static relation. It admits a lossless causal implementation with fresh Q -input at every step.

Lemma 1.3 (Lossless causal lift). *Suppose (3) holds at node h , and write $S_h = Q * B_h$. For $S_h(s) > 0$, define*

$$K_h(x \mid s) := \frac{P_x B_{hx}(s-x)}{S_h(s)}, \quad B_{hx}(k) := 0 \quad (k < 0).$$

If $J_t \sim B_h$, draw $Y_{t+1} \sim Q$ independently, then draw X_{t+1} according to $K_h(\cdot \mid J_t + Y_{t+1})$, and put

$$J_{t+1} = J_t + Y_{t+1} - X_{t+1}.$$

Then $J_{t+1} \geq 0$, the public output has law P , the fresh input remains independent of the past, and conditional on $X_{t+1} = x$ the new reserve has law B_{hx} .

Proof. For fixed $x, k \geq 0$,

$$\begin{aligned}\mathbb{P}(X_{t+1} = x, J_{t+1} = k) &= S_h(k+x)K_h(x \mid k+x) \\ &= P_x B_{hx}(k).\end{aligned}$$

Summing over k gives the P -marginal, while division by P_x gives the prescribed child law. The convention $B_{hx}(k) = 0$ for $k < 0$ forces nonnegative reserve. $\square$

Taking first moments in (3) and using $\mathbb{E}P = \mathbb{E}Q$ gives the exact fan conservation law

$$\boxed{\mathbb{E}B_h = \sum_{x=0}^4 P_x \mathbb{E}B_{hx}.} \quad (5)$$

Lemma 1.4 (Finite-support reduction). *For every finite horizon n , the infimum in (4) is attained, and an optimal root may be chosen with support in $\{0, \dots, 4n\}$.*

Proof. Use the causal realization above. For any feasible depth- n trajectory define

$$D = (J_0 - 4n)^+, \quad J'_t = J_t - D.$$

On $\{J_0 > 4n\}$,

$$J'_t \geq 4n - 4t \geq 0,$$

since one step can decrease the reserve by at most 4. Otherwise $D = 0$. The (X, Y) -trajectory is unchanged, so freshness and exactness are preserved, while the root becomes $\min(J_0, 4n)$. The resulting finite occupancy problem is a compact linear program. $\square$

2 A logarithmic lower bound

The quartic defect is small near $z = 1$, but it cannot be repaired with uniformly bounded expected reserve. The following argument is purely finite-horizon.

Theorem 2.1 (Logarithmic reserve growth). *Fix $0 < z < 1$ and write*

$$p := G_P(z), \quad q := G_Q(z), \quad c(z) := \frac{\log(p/q)}{-\log z} > 0.$$

For integers $r \geq t \geq 1$,

$$\boxed{\mathcal{B}_r \geq (1 - p^t)\mathcal{B}_{r-t} + t c(z)p^t.} \quad (6)$$

Consequently

$$\boxed{\mathcal{B}_n \geq \max_{1 \leq t \leq n} t c(z) \left[1 - (1 - p^t)^{\lfloor n/t \rfloor}\right],} \quad (7)$$

and

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\log(G_P(z)/G_Q(z))}{(-\log z)(-\log G_P(z))} > 0.} \quad (8)$$

Proof. Take an optimal depth- r tree. Put

$$F_h(z) := \mathbb{E}z^{J_h}, \quad m_h := \mathbb{E}B_h, \quad b := \mathcal{B}_{r-t}, \quad a := -\log z.$$

Iterating (3) through t levels gives

$$q^t F_\emptyset(z) = p^t \mathbb{E}_{P_z^{\otimes t}} F_g(z), \quad P_z(x) := \frac{P_x z^x}{p}.$$

Jensen's inequality yields $F_g(z) \geq e^{-am_g}$, hence

$$\mathbb{E}_{P_z^{\otimes t}} m_g \geq \frac{t \log(p/q) - \log F_\emptyset(z)}{a} \geq tc(z).$$

Every depth- t descendant is the root of a feasible depth- $(r-t)$ subtree, so $m_g \geq b$. Moreover

$$\frac{P_z^{\otimes t}(g)}{P(g)} = \frac{z^{\sum_i g_i}}{p^t} \leq p^{-t}.$$

Using (5),

$$\begin{aligned} \mathbb{E}_{P_z^{\otimes t}} m_g &= b + \mathbb{E}_{P_z^{\otimes t}} (m_g - b) \\ &\leq b + p^{-t} \sum_{|g|=t} P(g)(m_g - b) \\ &= b + p^{-t}(\mathcal{B}_r - b). \end{aligned}$$

Combining the last two displays gives (6). Iteration over blocks gives (7). Finally choose

$$t_n = \left\lfloor \frac{\log n - 2 \log \log n}{-\log p} \right\rfloor.$$

Then $t_n / \log n \rightarrow 1/(-\log p)$ and $\lfloor n/t_n \rfloor p^{t_n} \rightarrow \infty$, so the bracket in (7) tends to one. This gives (8). $\square$

At $z = 1/16$,

$$G_P(z) = \frac{74273}{524288}, \quad G_Q(z) = \frac{543559}{4194304},$$

so

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \kappa_0 := \frac{\log(594184/543559)}{\log 16 \log(524288/74273)} = 0.0164347253 \dots \quad (9)$$

3 Weak- L^1 reduction of the logarithmic upper bound

The matching upper bound can be reduced to an infinite exact tree whose root has critical $1/j$ tail.

Theorem 3.1 (Truncated-reserve criterion). *Allow an infinite exact tree whose node laws may have infinite first moment. If $J_0 \sim B_\emptyset$, then for every $n \geq 1$,*

$$\boxed{\mathcal{B}_n \leq \mathbb{E} \min(J_0, 4n) = \sum_{j=1}^{4n} \mathbb{P}(J_0 \geq j).} \quad (10)$$

If

$$\mathbb{P}(J_0 \geq j) \leq \frac{A}{j} \quad (j \geq 1),$$

then

$$\boxed{\mathcal{B}_n \leq AH_{4n} \leq A(1 + \log(4n)).} \quad (11)$$

Proof. For a fixed horizon n , subtract the same random quantity

$$D = (J_0 - 4n)^+$$

from every reserve during the first n steps. If $J_0 > 4n$, then

$$J'_t = 4n + \sum_{i=1}^t (Y_i - X_i) \geq 4n - 4t \geq 0.$$

The public and fresh-input trajectories are unchanged, so the shifted laws form an exact depth- n tree with root $\min(J_0, 4n)$. The rest is the tail-sum identity and the harmonic bound. $\square$

A convenient critical root for the packet construction below is

$$H_2(0) = 0, \quad H_2(j) = \frac{2}{(j+1)(j+2)} \quad (j \geq 1), \quad (12)$$

for which

$$\boxed{\mathbb{P}_{H_2}(J \geq j) = \frac{2}{j+1} \leq \frac{2}{j}.} \quad (13)$$

Thus an infinite exact tree rooted at H_2 would imply

$$\boxed{\mathcal{B}_n \leq 2H_{4n} = 2 \log n + O(1).} \quad (14)$$

4 One-step transport and exact tail interlacing

Let B be a parent reserve law and put $S = Y + J$, with $Y \sim Q$, $J \sim B$ independent.

Theorem 4.1 (One-fan equivalence). *A family of probability laws B_x satisfies*

$$Q * B = \sum_{x=0}^4 P_x \delta_x * B_x$$

*if and only if there exists a coupling π of $X \sim P$ and $S \sim Q * B$, supported on $X \leq S$, such that*

$$B_x(j) = \frac{\pi(x, x+j)}{P_x}.$$

Consequently a positive one-fan exists if and only if

$$\boxed{P \preceq_{\text{st}} Q * B.} \quad (15)$$

For (1), writing $b_j = B(j)$, this is equivalent to

$$27b_0 + 7b_1 \leq 24, \quad (16)$$

$$37b_0 + 27b_1 + 7b_2 \leq 40, \quad (17)$$

$$57b_0 + 37b_1 + 27b_2 + 7b_3 \leq 56. \quad (18)$$

Proof. From a fan define $\pi(x, s) = P_x B_x(s - x)$. Conversely, recover B_x from the displayed formula. The order support is exactly nonnegativity of the residual reserve. In one dimension, existence of an order coupling is equivalent to stochastic domination. Since P is supported on $\{0, 1, 2, 3, 4\}$, only the cumulative inequalities at $0, 1, 2, 3$ matter; direct substitution gives (16)–(18). See, e.g., [1, 2]. $\square$

The next theorem separates the infinite tail from the finite boundary transport.

Theorem 4.2 (Strict-core exact-tail interlacing). *Let $\mu = Q * B$ and suppose*

$$F_\mu(k) < F_P(k), \quad k = 0, 1, 2, 3.$$

Then there exists a finite cutoff $H \geq 4$ and an order coupling π such that

$$\boxed{\pi(x, s) = P_x \mu_s \quad (s > H).} \quad (19)$$

Hence every child obeys

$$\boxed{B_x(j) = \mu_{x+j} \quad (x+j > H).} \quad (20)$$

If

$$\mathbb{P}_B(J \geq j) = \frac{A}{j} + O(j^{-2}),$$

then for every x ,

$$\mathbb{P}_{B_x}(J \geq j) = \frac{A}{j} + O(j^{-2}). \quad (21)$$

Proof. Let $T_H = \mu((H, \infty))$ and $c_H = 1 - T_H$. Normalize the truncation by

$$\mu^{(H)}(s) = \frac{\mu_s}{c_H}, \quad 0 \leq s \leq H.$$

For $k = 0, 1, 2, 3$,

$$F_{\mu^{(H)}}(k) = \frac{F_\mu(k)}{c_H} \longrightarrow F_\mu(k) < F_P(k).$$

For all sufficiently large H , $P \preceq_{\text{st}} \mu^{(H)}$. Let ρ_H be an order coupling of these finite laws and set

$$\pi(x, s) = \begin{cases} c_H \rho_H(x, s), & s \leq H, \\ P_x \mu_s, & s > H. \end{cases}$$

This has the required marginals and support. Equation (20) follows immediately. Since Y has bounded support, convolution with Q preserves a tail $A/j + O(j^{-2})$, and a bounded shift by x preserves the same leading coefficient. $\square$

5 A positive three-regime packet fan

We now give an explicit causal coupling that isolates the hard boundary correction into a selectable positive packet.

For a measure M on $\mathbb{N}_0$, define

$$(DM)(j) := M(j+1), \quad (UM)(j) := \begin{cases} 0, & j = 0, \\ M(j-1), & j \geq 1. \end{cases}$$

Consider the three Q -column couplings

$$C^+ = \frac{1}{64} \begin{pmatrix} 7 & 4 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 & 0 \\ 0 & 0 & 10 & 4 & 2 \\ 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}, \quad (22)$$

$$C^0 = \frac{1}{64} \begin{pmatrix} 7 & 1 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 & 0 \\ 0 & 3 & 10 & 3 & 0 \\ 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 1 & 7 \end{pmatrix}, \quad (23)$$

and

$$C^- = \frac{1}{64} \begin{pmatrix} 7 & 0 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 & 0 \\ 0 & 4 & 10 & 2 & 0 \\ 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 2 & 7 \end{pmatrix}. \quad (24)$$

Their row laws are

$$R^+ = P + \frac{3}{64}(\delta_0 - \delta_4), \quad R^0 = P, \quad R^- = P - \frac{1}{64}(\delta_0 - \delta_4). \quad (25)$$

Moreover C^+ is supported on $X \leq Y$, while C^0, C^- are supported on $X \leq Y + 1$.

Theorem 5.1 (Packet fan). *Let B be a probability law on $\mathbb{N}_0$, let $p = B(0) \leq 1/4$, and choose any submeasure*

$$G \leq B - p\delta_0, \quad |G| = 3p. \quad (26)$$

Put

$$A := B - p\delta_0 - G.$$

Use C^+ on the boundary mass $p\delta_0$, C^- on G , and C^0 on A . Then the resulting one-step protocol is causal, has fresh input law Q , public output law exactly P , and nonnegative residual reserve.

Its five child laws are

$$B_0 = \frac{1}{8} [p(7\delta_0 + 4\delta_1) + 7A + \mathbf{U}A + 7G], \quad (27)$$

$$B_1 = B, \quad (28)$$

$$B_2 = \frac{1}{16} [p(10\delta_0 + 4\delta_1 + 2\delta_2) + 3DA + 10A + 3\mathbf{U}A + 4DG + 10G + 2UG], \quad (29)$$

$$B_3 = B, \quad (30)$$

$$B_4 = \frac{1}{8} [5p\delta_0 + DA + 7A + 2DG + 7G]. \quad (31)$$

Proof. The public row law is

$$pR^+ + |G|R^- + |A|P.$$

Using $|G| = 3p$, $|A| = 1 - 4p$, and (25), the correction in $\delta_0 - \delta_4$ cancels exactly, leaving P . At $J = 0$, C^+ never spends more than the fresh input. On $J \geq 1$, both C^0 and C^- satisfy $X \leq Y + 1$, so one unit of reserve is sufficient. Reading each row and converting $j \mapsto j + Y - X$ gives (27)–(31). In particular the rows $x = 1, 3$ are exact identities in all three couplings. $\square$

The identities

$$\boxed{B_1 = B_3 = B} \quad (32)$$

are central: the two odd public branches transport the complete hidden state without distortion.

6 Top-tail packet selection and exact prefix dynamics

The packet location is a genuine phase degree of freedom. A particularly sharp deterministic choice is to take G from the highest reserve states downward, splitting the threshold atom if needed.

Write

$$p = B(0), \quad q = B(1), \quad r = B(2).$$

For the top-tail packet of mass $3p$, define

$$g_1 := G(1) = (4p + q - 1)_+, \quad (33)$$

and

$$g_2 := G(2) = (4p + q + r - 1)_+ - (4p + q - 1)_+. \quad (34)$$

Proposition 6.1 (Exact low-prefix recursion). *For the top-tail packet fan,*

$$p_0 := B_0(0) = \frac{7}{8}p, \quad q_0 := B_0(1) = \frac{1}{2}p + \frac{7}{8}q, \quad (35)$$

$$p_1 = p_3 = p, \quad q_1 = q_3 = q, \quad (36)$$

$$p_2 = \frac{5}{8}p + \frac{3}{16}q + \frac{1}{16}g_1, \quad q_2 = \frac{1}{4}p + \frac{10}{16}q + \frac{3}{16}r + \frac{1}{16}g_2, \quad (37)$$

$$p_4 = \frac{5}{8}p + \frac{1}{8}q + \frac{1}{8}g_1, \quad q_4 = \frac{7}{8}q + \frac{1}{8}r + \frac{1}{8}g_2. \quad (38)$$

Proof. Equations (33)–(34) are the defining cumulative identities of the highest-mass submeasure of total mass $3p$: the packet reaches state 1 only after exhausting all mass above 1, and similarly for state 2. The formulas then follow by reading the coefficients of δ_0 and δ_1 in (27)–(31). $\square$

The first boundary face collapses to a single linear inequality.

Theorem 6.2 (Sharp one-step boundary criterion for the top-tail packet). *Assume $p \leq 1/4$. Under the top-tail packet fan,*

$$\boxed{B_x(0) \leq \frac{1}{4} \quad \text{for every } x = 0, \dots, 4} \quad (39)$$

if and only if

$$\boxed{14p + 4q \leq 5.} \quad (40)$$

Proof. Branches 1, 3 preserve p , and branch 0 maps p to $7p/8$. Moreover $p_4 \leq p_2$, so only branch 2 can be critical.

If $4p + q \leq 1$, then $g_1 = 0$ and

$$p_2 = \frac{10p + 3q}{16} < \frac{1}{4}$$

under the same constraint. If $4p + q > 1$, then $g_1 = 4p + q - 1$, so

$$p_2 = \frac{14p + 4q - 1}{16}.$$

Thus $p_2 \leq 1/4$ is exactly (40). In the first region (40) holds automatically, so the equivalence is global. $\square$

If one asks that the children themselves satisfy (40), only the active branches 0, 2, 4 matter. Direct substitution gives the next three exact conditions:

$$57p + 14q \leq 20, \quad (41)$$

$$78p + 41q + 6r + 7g_1 + 2g_2 \leq 40, \quad (42)$$

$$70p + 42q + 4r + 14g_1 + 4g_2 \leq 40. \quad (43)$$

This exposes the recursive problem as an explicit prefix hierarchy rather than an implicit transport linear program.

7 Waring/Beta phase carriers

The physical top-tail choice is sharp for one step, but recursive closure benefits from a basis that remembers how mass arrived at the boundary. The following family provides such a phase coordinate.

Definition 7.1 (Waring-type phase family). For $k \geq 0$, let V_k be the positive-integer-valued law with

$$\mathbb{P}(V_k \geq j) = \frac{k+1}{k+j}, \quad j \geq 1, \quad (44)$$

and define

$$W_{r,k} := \text{Law}(r + V_k), \quad r \geq 0.$$

Lemma 7.2 (Waring aging identity). *For every $k \geq 0$,*

$$\boxed{\text{Law}(V_k - 1) = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}V_{k+1}.} \quad (45)$$

Consequently

$$\boxed{DW_{0,k} = \frac{1}{k+2}\delta_0 + \frac{k+1}{k+2}W_{0,k+1},} \quad (46)$$

while for $r \geq 1$,

$$DW_{r,k} = W_{r-1,k}, \quad UW_{r,k} = W_{r+1,k}. \quad (47)$$

Proof. For $j \geq 1$,

$$\mathbb{P}(V_k - 1 \geq j) = \frac{k+1}{k+j+1} = \frac{k+1}{k+2} \frac{k+2}{k+j+1}.$$

The missing mass is $1/(k+2)$ at zero. The shifted identities are immediate. $\square$

There is also a continuous Beta-mixture representation. Let

$$g_t(j) = (1-t)t^{j-1}, \quad j \geq 1,$$

and $E_{r,t} := \delta_r * g_t$. Then

$$\boxed{W_{r,k} = (k+1) \int_0^1 t^k E_{r,t} dt.} \quad (48)$$

Indeed $\mathbb{P}(E_{r,t} \geq r+j) = t^{j-1}$, and integration gives (44). At the boundary,

$$\boxed{DE_{0,t} = (1-t)\delta_0 + tE_{0,t},} \quad (49)$$

which makes the phase interpretation transparent: surviving a boundary hit multiplies the continuous component by t , while the leaked mass $1-t$ becomes boundary mass.

The critical root (12) is exactly

$$\boxed{H_2 = W_{0,1}.} \quad (50)$$

Definition 7.3 (Waring packet cone). Let $\mathfrak{W}$ be the cone of finite positive measures representable as

$$M = \sum_{j \geq 0} d_j \delta_j + \sum_{r,k \geq 0} w_{r,k} W_{r,k}, \quad d_j, w_{r,k} \geq 0, \quad (51)$$

with countable sums interpreted by monotone convergence whenever the total mass is finite. A *componentwise submeasure* is obtained by decreasing the coefficients $d_j, w_{r,k}$ without changing the component shapes.

Proposition 7.4 (Positive phase closure of the packet fan). *Suppose $B \in \mathfrak{W}$ is a probability law, $p = B(0) \leq 1/4$, and the packet G in Theorem 5.1 is chosen componentwise from a representation (51). Then every child B_x belongs to $\mathfrak{W}$.*

Proof. Equations (27)–(31) express each child as a positive linear combination of atoms and of A, G after applying only I, D, U . Equations (46)–(47) show that these operators preserve $\mathfrak{W}$. The explicit safe contribution from $p\delta_0$ produces only $\delta_0, \delta_1, \delta_2$, which already lie in the cone. $\square$

7.1 Phase leak and greedy component selection

For a Waring component, the amount of mass sitting at reserve one is explicit:

$$W_{0,k}(1) = \frac{1}{k+2}, \quad W_{r,k}(1) = 0 \quad (r \geq 1). \quad (52)$$

Thus the phase index k has an exact operational meaning: after a boundary down-shift, a surviving component ages from k to $k+1$, and its next boundary leak decreases.

Given a representation (51), a componentwise packet of prescribed mass can be chosen by fractional knapsack: first use components with smaller reserve-one leakage, splitting the last coefficient if necessary. This minimizes $G(1)$ among packets that respect the chosen phase decomposition.

Lemma 7.5 (One-step leak bound). *Let $B \in \mathfrak{W}$ be a probability law, $p = B(0) \leq 1/4$, and $q = B(1)$. There exists a componentwise packet $G \leq B - p\delta_0$ of mass $3p$ such that*

$$\boxed{G(1) \leq \frac{3p}{1-p} q.} \quad (53)$$

Proof. The positive part has total mass $1-p$. The proportional componentwise packet

$$G_{\text{prop}} = \frac{3p}{1-p} (B - p\delta_0)$$

has mass $3p$ and satisfies equality in (53). A packet minimizing the reserve-one leakage over the same component representation can only improve this value. $\square$

Let

$$\eta := \frac{q}{1-p}.$$

Combining (53) with the exact packet formulas gives

$$B_2(0) \leq \frac{10p + 3\eta}{16}, \quad (54)$$

$$B_4(0) \leq \frac{5p}{8} + \frac{1+2p}{8}\eta. \quad (55)$$

Hence the boundary face

$$\boxed{0 \leq p \leq \frac{1}{4}, \quad 0 \leq \eta \leq \min\left(1, \frac{4-10p}{3}\right)} \quad (56)$$

is sufficient for one-step control of the zero mass in all active children. Indeed, (54) is at most $1/4$ under the second inequality in (56). For (55), the required bound is

$$\eta \leq \frac{2-5p}{1+2p}.$$

On $0 \leq p \leq 1/4$,

$$\frac{4-10p}{3} \leq \frac{2-5p}{1+2p},$$

because the difference after cross multiplication is

$$2 - 13p + 20p^2 = (4p-1)(5p-2) \geq 0.$$

For branch 0, the update is independent of the packet placement:

$$p_0 = \frac{7}{8}p, \quad \eta_0 = \frac{4p + 7\eta(1-p)}{8-7p}. \quad (57)$$

Moreover branch 0 preserves the face (56). To see this, note that η_0 is increasing in η . On the sloping boundary $\eta = (4 - 10p)/3$, the desired inequality

$$\eta_0 \leq \frac{4 - 10p_0}{3}$$

is equivalent to

$$35p^2 + 48p - 16 \leq 0.$$

Its positive root is $4(-6 + \sqrt{71})/35 > 1/4$, so it holds throughout $[0, 1/4]$. On the cap $\eta = 1$,

$$\eta_0 = \frac{7-3p}{8-7p} \leq 1.$$

What remains is to use the Waring phase freedom to control the corresponding normalized low-prefix quantities on branches 2 and 4 simultaneously through all histories.

8 The remaining Waring packet invariance problem

We now isolate the single statement that would complete the logarithmic construction developed in this preprint.

Conjecture 8.1 (Waring packet invariance). *Start from*

$$B_\emptyset = H_2 = W_{0,1}.$$

At every reachable node h , write $p_h = B_h(0)$. There exists a componentwise packet

$$G_h \leq B_h - p_h \delta_0, \quad |G_h| = 3p_h,$$

chosen from a Waring representation (51), such that the packet fan (27)–(31) can be iterated for every finite history and satisfies

$$\boxed{B_h(0) \leq \frac{1}{4} \quad \text{for every reachable history } h.} \quad (58)$$

Equivalently, the componentwise phase inventory can always be selected so that the boundary compensation capacity $|G_h| = 3B_h(0)$ regenerates indefinitely.

The conjecture is deliberately formulated at the level actually used by the positive construction. It does not require a stationary child law, a fixed physical tail cutoff, or a history-independent scalar feedback. The state is the positive Waring phase inventory itself. The exact aging identity (46) is the mechanism that can move mass away from the hard boundary without leaving the positive cone.

Theorem 8.2 (Conditional logarithmic upper bound). *If the Waring packet invariance conjecture holds, then an infinite exact zero-repair tree exists with root H_2 , and*

$$\boxed{\mathcal{B}_n \leq 2H_{4n} = 2 \log n + O(1).} \quad (59)$$

Combined with (9), this gives

$$\boxed{\mathcal{B}_n = \Theta(\log n).} \quad (60)$$

Proof. The conjecture recursively supplies a valid positive packet fan at every output history. By the causal lift, this defines an infinite exact tree with fresh iid Q -inputs and iid P -outputs. The root tail is (13); Theorem 3.1 then gives (59). The lower bound follows from (9). $\square$

9 Discussion

The constructive picture is now unusually explicit. The global logarithmic question has been reduced to a positive phase-selection problem inside a shift-stable cone:

1. the hard boundary creates a public-law defect of size $3p$;
2. a packet of positive reserve mass $3p$ cancels that defect exactly;
3. branches 1 and 3 are exact identity channels for the complete hidden law;
4. the top-tail packet exposes exact low-prefix recursions, including the sharp one-step criterion $14p + 4q \leq 5$;
5. the Waring basis turns a boundary down-shift into a positive aging operation, with leak $1/(k+2)$ and survivor phase $k+1$;
6. the critical root $H_2 = W_{0,1}$ already has the weak- L^1 tail required for the logarithmic bill.

The missing theorem is therefore not an asymptotic estimate and not an existence statement for a generic transport polytope. It is the all-history regeneration of enough low-leak Waring phase mass to keep the packet fan inside the boundary-capacity region. Establishing that invariant, or finding the smallest additional phase statistic required to state it correctly, is the next mathematical step.

A Exact matrix audit for the packet fan

For completeness, we record the elementary identities behind Section 5. Multiplying by 64, the column sums of each of C^+ , C^0 , C^- are

$$(7, 20, 10, 20, 7) = 64Q.$$

The row sums are respectively

$$(11, 16, 16, 16, 5), \quad (8, 16, 16, 16, 8), \quad (7, 16, 16, 16, 9),$$

which are exactly $64R^+$, $64P$, $64R^-$. Thus

$$pR^+ + 3pR^- + (1 - 4p)P = P.$$

All nonzero cells of C^+ satisfy $x \leq y$, and all nonzero cells of C^0, C^- satisfy $x \leq y + 1$. Hence the construction is solvent on $J = 0$ and on $J \geq 1$, respectively.

B Notation cross-check

The Waring root $H_2 = W_{0,1}$ has

$$\mathbb{P}(H_2 \geq j) = \frac{2}{j+1}, \quad H_2(j) = \frac{2}{(j+1)(j+2)}, \quad j \geq 1,$$

and no atom at zero. Its truncated first moment is therefore

$$\mathbb{E} \min(H_2, N) = \sum_{j=1}^N \frac{2}{j+1} = 2(H_{N+1} - 1).$$

This is the exact quantity underlying the conditional estimate (59).

References

- [1] V. Strassen, The existence of probability measures with given marginals, *Annals of Mathematical Statistics* **36** (1965), 423–439.
- [2] M. Shaked and J. G. Shanthikumar, *Stochastic Orders*, Springer Series in Statistics, Springer, 2007.